

CADEMAS-ML: A Software Tool for Context-Aware Cooperative and Explainable Automated Decision-Making

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ARTICLE HISTORY

Compiled October 2, 2026

ABSTRACT

Complex organizational decisions often combine assessments from multiple actors with the context surrounding the problem. The Cooperative Automated Decision-Making Systems (CADEMAS) framework was recently proposed to represent this form of cooperative and context-aware decision-making. However, previous CADEMAs instantiations cover only specific configurations of a framework that allows substantially broader combinations of automated decision components, contextual models, and aggregation mechanisms. This breadth makes the framework difficult to operationalize without dedicated software support. This paper presents CADEMAs-ML, a software tool that implements a configurable class of CADEMAs systems in which automated decision-making components are machine learning models and contextual knowledge is represented through fuzzy logic. It integrates weighted predictive models with configurable fuzzy contexts and alternative aggregation operators. Explainability and robustness analyses support inspection of the resulting decisions. An employee attrition case study illustrates how the same predictive evidence can lead to different prioritizations under alternative contextual scenarios.

KEYWORDS

automated decision-making systems; fuzzy decision making; context-aware decision support; machine learning; prioritization; cooperative systems

1. Introduction

Decision-making in real-world organizations is rarely a matter of applying a single analytical model to a well-defined set of inputs. Complex decisions often involve multiple stakeholders, organizational units, or analytical components that contribute different pieces of evidence to the same problem. Their objectives, information, expertise, and

reliability may differ. Coordination is therefore needed before the evidence can support a coherent course of action.

This perspective becomes particularly important when automated decision-making is used to support prediction, classification, or risk assessment. An automated decision-making (ADM) component may produce a probability, rating, recommendation, or risk score. Acting on that output also requires attention to organizational priorities, available resources, regulations, and strategy. Decisions are thus made within a context.

Several ADM components can provide complementary views of the same cases. Domain knowledge and organizational policies then guide how those views inform the final decision.

These considerations motivated the Cooperative Automated Decision-Making Systems (CADEMAS) framework proposed by Novoa-Hernández, Pelta, Godz, Verdegay, and Buendia-Carrillo (2026). CADEMAs coordinates heterogeneous ADM components and incorporates the context in which their outputs are interpreted. It connects an integrated assessment with a separate representation of organizational priorities, policies, or scenarios. Section 2.1 formalizes this architecture.

The framework builds on cooperative and multi-agent decision-support systems, where heterogeneous components contribute to a common process (Bui & Lee, 1999; Vahidov & Fazlollahi, 2004). It also draws on multiperson and group decision support, which combines heterogeneous information and preferences through aggregation, negotiation, and consensus mechanisms (Herrera-Viedma, Herrera, & Chiclana, 2002; Salo, 1995). In an automated setting, software-based decision units assume the cooperating roles.

One relevant scenario involves a set of entities and a limited intervention budget. The decision problem is to establish which entities should receive attention first.

Examples include attrition risk prevention Godz, Novoa-Hernández, and Pelta (2026), financial rescue for small and medium-sized companies Navarro-Galera, Lara-Rubio, Novoa-Hernández, and Cruz Corona (2025), student-dropout prevention (BUSCAR REF).

In these problems, CADEMAs combines the assessments of multiple ADMs with the suitability of each entity for the decision context. Section 3 develops this connection for prioritization. Fuzzy rules can represent such contexts (Lamata, Pelta, & Verdegay, 2020; Pérez-Cañedo, Porras, Pelta, & Verdegay, 2023), combining quantitative information with expert knowledge and linguistic assessments.

Users also need to understand individual predictions, especially when they influence organizational or human outcomes (Barredo Arrieta et al., 2020). Local feature attributions can reveal the factors associated with each prediction (Lundberg et al., 2020; Lundberg & Lee, 2017).

The ordering of cases may also change with the contextual definition, aggregation mechanism, or balance assigned to predictive and contextual information. Robustness

analysis reveals stable priorities and cases that are sensitive to these policy choices (Barredo Arrieta et al., 2020).

To the best of our knowledge, no integrated tool supports this complete prioritization workflow.

This paper presents CADEMAs-ML to fill that gap. The tool implements a configurable class of CADEMAs systems for prioritization, with machine-learning ADM units and fuzzy contextual knowledge. It also extends the workflow with case-level explanation and robustness analysis.

The remainder of the paper is organized as follows. Section 2 reviews the CADEMAs framework and related work on cooperative and fuzzy decision-support systems. Section 3 formalizes its specialization for prioritization. Section 4 presents the software architecture and its main functionalities through an employee-attrition case study. Finally, Section 5 summarizes the main findings and discusses the implications of the proposed tool and possible directions for future work.

2. Background and related works

This section sets out the conceptual foundations on which CADEMAs-ML builds. We first revisit the CADEMAs framework in its general form (Section 2.1). We then situate the tool among related cooperative-decision frameworks and application-oriented decision-support systems, clarifying the gap it is designed to fill (Section 2.2).

2.1. CADEMAs framework

As introduced in Section 1, CADEMAs (Novoa-Hernández et al., 2026) provides a general conceptual and architectural framework for structuring cooperation among heterogeneous Automated Decision-Making (ADM) units. Its scope is not restricted to prediction. The participating components may be classifiers, optimization or simulation models, rule-based systems, software agents, or institutional procedures. Their assumptions, data, outputs, and reliability may differ. CADEMAs defines how they contribute to a common process and how context informs the resulting decision without imposing a particular learning method or coordination protocol.

An ADM unit may contribute a *partial decision signal*, such as a score, prediction, rule activation, explanation, or suggested action. Signals with different semantics are integrated and then evaluated in their context. Formally, a CADEMAs instance is represented by the tuple

$$\mathcal{S} = \langle \mathcal{IM}, \mathcal{A}, \mathcal{OM}, \mathcal{C}, \mathcal{T} \rangle, \quad (1)$$

where $\mathcal{IM}$ is the Input Manager, $\mathcal{A}$ is the set of ADM units, $\mathcal{OM}$ is the Decision

Integration Layer, $\mathcal{C}$ is the Contextual Modulator, and $\mathcal{T}$ is the universe of admissible data types and representations.

The Input Manager $\mathcal{IM}$ routes, transforms, and validates the information entering the architecture. It distinguishes training or calibration data, which are used offline to construct or update the internal models of the ADM units, from decision-case data provided when a new case is evaluated. The set of ADM units is denoted by

$$\mathcal{A} = \{ADM_1, \dots, ADM_n\}. \quad (2)$$

Each ADM_i receives a data representation $d_i \in \mathcal{T}$ and produces an output $o_i \in \mathcal{T}$. Cooperation among these units may be *indirect*, with each unit contributing an independent signal, or *direct*, with units exchanging intermediate representations. CADEMAs allows for both forms without imposing a specific communication protocol.

Given the heterogeneous collection $\mathcal{O} = \{o_1, \dots, o_n\}$, the Decision Integration Layer $\mathcal{OM}$ synthesizes a unified decision object D through a mapping

$$\Phi : \mathcal{O} \rightarrow D.$$

This layer addresses the harmonization of ADM outputs and their subsequent aggregation or synthesis. The resulting D is a *raw* proposal in a common representation. The Contextual Modulator then adjusts it to the relevant conditions. Formally,

$$M : D \times \mathcal{C} \rightarrow D, \quad D^{\text{final}} = M(D^{\text{raw}}, \mathcal{C}), \quad (3)$$

where $\mathcal{C}$ may encode regulatory requirements, organizational policies, resource constraints, or other scenario-dependent conditions. Its representation may use rules, utility functions, constraints, fuzzy propositions, or other mechanisms suited to the application.

The predictive and fuzzy specialization implemented by CADEMAs-ML, including cooperative risk scores, fuzzy context alignment, and an explicit trade-off between these components, is described in Section 3. The general architecture leaves the ADM types, output semantics, integration mapping Φ , and contextual modulation M open to specialization.

This flexibility is reflected in the applications reported to date. The original formulation illustrated the framework with an SME insolvency-prediction scenario in which heterogeneous decision units were combined with contextual constraints (Novoa-Hernández et al., 2026). Godz et al. (2026) subsequently instantiated CADEMAs for employee-attrition prevention, treating departmental machine-learning models as co-operating ADM units and encoding organizational scenarios through a fuzzy context layer. Building on this machine-learning pathway, Novoa-Hernández, Godz, and Pelta

(2026) examined the effect of different aggregation operators on prioritization, comparing linear, weighted-geometric, and minimum operators and analyzing how their aggregation behavior affects the interaction between predictive evidence and contextual requirements. More recently, Vazquez-Sánchez, Buendía-Carrillo, and Corona (2026) proposed a cooperative ADM strategy with fuzzy contextual modulation for SME insolvency prediction, combining heterogeneous survival models, a contextual modulator, and explainable-AI support.

Together, these studies show the range of possible CADEMAs configurations. CADEMAs-ML turns their shared machine-learning pathway into a reusable environment for constructing, inspecting, comparing, and reproducing such configurations.

2.2. *Related frameworks and decision-support applications*

Recent research has increasingly recognized that effective decision support requires more than accurate isolated models. In many application domains, decisions emerge from the interaction between automated units, expert knowledge, and contextual considerations that may change over time. For clarity, it is useful to distinguish two related but different bodies of work: general cooperative-decision frameworks, which address how multiple decision entities interact, and application-oriented decision-support systems, which implement concrete methods for a domain or family of decision problems.

At the framework level, cooperative game theory offers one formal perspective on how decision-making entities with heterogeneous roles or attributes may combine their contributions, including settings in which each agent is characterized by multiple attributes rather than a single payoff value (Borkotokey, Hazarika, & Mesiar, 2019). CADEMAs does not adopt this game-theoretic formulation, but shares its concern with making cooperation among heterogeneous entities explicit. At the operational level, Tariq, Chhetri, Nepal, and Paris (2024) propose an AI-to-collaborator (A2C) framework in which humans and AI systems share decision authority through explicit interaction mechanisms. Related work on hybrid intelligence emphasizes that authority distribution and human interpretive capacity should be modeled separately, because the level of AI integration and the cognitive flexibility of human decision-makers jointly shape decision quality and accountability (Deliu & Moldovan, 2026). A neighboring line of work examines cooperation among multiple AI agents, including LLM-based multi-agent reinforcement learning (Sun, Huang, & Pompili, 2024). These contributions provide relevant points of comparison because they address cooperation among decision entities. However, their main focus is not the reusable implementation of a CADEMAs-like workflow in which heterogeneous ADM outputs are explicitly separated from context modulation and final prioritization.

A related theoretical thread connects cooperative game theory to case-level explanation. The Shapley value distributes a joint outcome among players according to

their averaged marginal contributions (Borkotokey et al., 2019) and has been adopted in explainable artificial intelligence as a principled attribution mechanism. SHapley Additive exPlanations (SHAP) (Lundberg & Lee, 2017) treat model features as players and decompose each prediction into additive feature contributions with axiomatic guarantees; recent surveys document the breadth of this cooperative-to-XAI bridge (Li, Sun, Huang, & Chen, 2024). This literature motivates the inclusion of an explicit explanation module in CADEMAs-ML. The current implementation does not compute Shapley values over features: because the uploaded H2O MOJO models are treated as black boxes and may include stacked ensembles, the tool uses a model-agnostic perturbation procedure instead. The purpose is nonetheless the same as in Shapley-inspired XAI: to make the local drivers of a recommendation inspectable alongside the cooperative and contextual scores.

At the application and decision-support level, substantial progress has been made in fuzzy decision support, multi-criteria decision making, and hybrid fuzzy-ML workflows. Applications such as MAKEDECISION (Więckowski & Sałabun, 2024) and the fuzzy DEMATEL-based tool of Chekry et al. (2024) emphasize transparency and traceability by exposing intermediate calculations to users. A related stream has explored hybrid pipelines that combine expert judgment, causal weighting, and predictive components in domain-specific settings (Abdulgader, Eid, & Daneshvar Rouyendegh, 2018; Özkurt, Canay, Tunç, Aydın, & Velioğlu, 2025; Sahraei et al., 2022). Recent work has also extended the fuzzy toolbox through aggregation operators that account for interactions among decision attributes, capturing dependency structures that simple weighted averages cannot represent (Kacprzyk, Sirbiladze, & Tsulaia, 2022). The importance of explainability in AI-based decision support has been broadly recognized: any automated system whose outputs must be justified or audited requires transparent mechanisms to convey its reasoning to stakeholders (Barredo Arrieta et al., 2020), a requirement increasingly addressed through XAI techniques, including Shapley-value attribution (Li et al., 2024; Lundberg & Lee, 2017), in hybrid ranking and prediction pipelines (Özkurt et al., 2025). These contributions demonstrate the value of combining data-driven evidence with contextual or expert-defined criteria, but most are tailored to particular methods or domains rather than organized around the CADEMAs decomposition of ADM units, integration, context, and final prioritization.

Open-source fuzzy and MCDM libraries, including PYFDM (Więckowski, Kizielewicz, & Sałabun, 2022), PYIFDM (Więckowski, Kizielewicz, & Sałabun, 2023), the IFMCDM package of Roszkowska, Jefmański, Dudek, and Kusterka-Jefmańska (2024), the ELIGERE platform (Grazioso, Caporaso, & di Gironimo, 2023), and the intuitionistic fuzzy dimensional-analysis approach of Pérez-Domínguez, Callejas-Cuervo, and García-Luna (2024), are also relevant. They improve reproducibility and provide reusable computational building blocks.

Table 1 summarizes representative application-oriented contributions in these ar-

Table 1. Comparison of representative application-oriented decision-support works related to CADEMAs-ML.

Work	Decision-support focus	Multiple ADM units	Explicit context layer	Case-level XAI	Relation to CADEMAs-ML
Tariq et al. (2024)	Human–AI collaborative decision framework with automated, augmented, and collaborative modes	Partly	Partly	No	Strong framework-level precedent, but not a configurable fuzzy/ML prioritization workflow
Więckowski and Sałabun (2024)	Auditable fuzzy MCDM through a web-based tool	No	Partly	No	Strong transparency, but no cooperative ADM integration layer
Chekry et al. (2024)	Open-source fuzzy DEMATEL decision-support tool	No	Partly	No	Traceable fuzzy causal analysis, but method-specific rather than CADEMAs-like
Özkurt et al. (2025)	Hybrid fuzzy MCDM, ML, and XAI for transparent ranking and prediction	Partly	Partly	Partly	Combines ranking, prediction, and XAI, but lacks a general cooperative ADM architecture
Sahraei et al. (2022)	Fuzzy DEMATEL/BWM/ANP pipeline for spatial risk prioritization	No	Partly	No	Explicit causal weighting under uncertainty, but application-specific
Godz et al. (2026)	Employee-attribution prevention using CADEMAs with departmental ML models	Yes	Yes	No	Direct CADEMAs application that motivates the implemented workflow
Novoa-Hernández et al. (2026)	Aggregation semantics for CADEMAs prioritization under alternative operators	Yes	Yes	No	Analyzes how linear, geometric, and minimum rules preserve or compensate contextual requirements
Vazquez-Sánchez et al. (2026)	Cooperative ADM with fuzzy contextual modulation for SME insolvency prediction	Yes	Yes	Yes	Extends CADEMAs-style cooperation to survival models with contextual modulation and XAI support
CADEMAs-ML (this work)	Configurable prioritization with ML-based ADM units, fuzzy context scenarios, and local case explanation	Yes	Yes	Yes	Operationalizes CADEMAs for reusable, explainable ML-driven decision support

eas. Three observations emerge from this comparison. First, published systems already provide strong ingredients for transparent decision support, including human–AI collaboration, auditable fuzzy MCDM, hybrid ranking pipelines, and reusable software libraries (Özkurt et al., 2025; Tariq et al., 2024; Więckowski et al., 2022; Więckowski & Sałabun, 2024). Second, these ingredients are usually developed in isolation. Third, recent CADEMAs applications already combine cooperative multi-ADM integration

with an explicit contextual layer (Godz et al., 2026; Novoa-Hernández et al., 2026; Vazquez-Sánchez et al., 2026), yet they remain domain-specific configurations rather than a reusable software workflow with case-level explanation. What remains missing is an operational implementation that keeps these capabilities configurable and auditable together.

CADEMAS-ML is designed to address this gap for the predictive-model setting. Building on the framework and its attrition instantiation discussed above, it provides a configurable software implementation in which ML-based ADM outputs, contextual reasoning, and case-level explanation coexist as explicit and auditable components of the decision-making process. The source code is publicly available under an MIT license (Section 5); the main contribution is the operationalization of the CADEMAs decomposition in a reusable decision-support tool.

3. Instantiation of CADEMAs for the Prioritization Problem

3.1. Problem definition and overall mapping

Section 1 motivates prioritization when attention, intervention, or resources are limited. This section now formalizes that problem. Let $\mathcal{X} = \{x_1, \dots, x_N\}$ denote the set of cases considered in a decision process, where each case x_i is characterized by a set of observable attributes. The objective is to score the cases and establish a priority ordering over $\mathcal{X}$.

Let

$$P : \mathcal{X} \rightarrow [0, 1] \tag{4}$$

denote a priority function that assigns a normalized priority score to each case. This score induces a preference relation over the cases such that

$$x_i \succeq_P x_j \iff P(x_i) \geq P(x_j). \tag{5}$$

The corresponding ranking can be represented by a function

$$\pi_P : \mathcal{X} \rightarrow \{1, \dots, N\}, \tag{6}$$

where lower values indicate higher priority. In the absence of ties, the rank of a case is therefore

$$\pi_P(x_i) = 1 + |\{x_j \in \mathcal{X} : P(x_j) > P(x_i)\}|. \tag{7}$$

Thus, a prioritization configuration defines both a priority score for each case and an ordering of the cases induced by those scores.

A priority function may draw on several assessments and produce different rankings under different organizational conditions. The integration and modulation stages defined in Section 2.1 provide the required structure. CADEMAs-ML specializes this architecture for the prioritization problem by implementing the ADM units as machine-learning models and providing explicit mechanisms for defining the data, model, and contextual components involved in the decision process. The resulting specialization can be summarized through the mapping

$$\mathcal{S} = \langle \mathcal{IM}, \mathcal{A}, \mathcal{OM}, \mathcal{C}, \mathcal{T} \rangle$$

as follows:

- **Data representation ($\mathcal{T}$):** Tabular case data provide the variables and decision signals required by the predictive and contextual components. Each component may use its own subset.
- **Input Manager ($\mathcal{IM}$):** The Input Manager coordinates case data, model requirements, performance metadata, and contextual definitions. These inputs can change without modifying the decision process.
- **ADM units ($\mathcal{A}$):** Pre-trained machine-learning models provide predictive perspectives on each case. A model may use its own attribute subset and contributes one partial signal.
- **Decision Integration Layer ($\mathcal{OM}$):** The integration layer combines the model assessments into a predictive score. Reported performance determines each model's contribution.
- **Contextual Modulator ($\mathcal{C}$):** Context definitions select relevant case attributes and evaluate their suitability for a scenario. The modulator combines this assessment with the predictive score.

The following subsections formalize these components and then introduce the explainability and robustness extensions.

3.2. *Input Manager, ADM Units and Data Representation*

The Input Manager $\mathcal{IM}$ groups the information needed to configure and execute a prioritization. It is represented as the tuple

$$\mathcal{IM} = \langle \mathcal{M}, \mathcal{M}_{def}, \mathcal{C}_{def}, \mathcal{X} \rangle, \tag{8}$$

where:

- $\mathcal{M} = M_1, \dots, M_K$ is the collection of pre-trained machine-learning models used as ADM units. Each model provides a partial predictive assessment of the cases and may rely on a different subset of the available features.
- $\mathcal{M}_{def} = m_1, \dots, m_K$ contains the corresponding model definitions. Each definition records model identity, predictive performance, and input features.
- $\mathcal{C}_{def}$ is the collection of contextual definitions used to configure the contextual component of the instance. Each definition identifies scenario-relevant attributes and their suitability rules.
- $\mathcal{X} = x_1, \dots, x_N$ is the set of cases to be prioritized. Each case is represented by a set of observable features sufficient to satisfy the input requirements of the configured models and contextual definitions, i.e.,

$$f(x_i) \supseteq \left(\bigcup_{k=1}^K f_k \right) \cup f_C.$$

where f_k denotes the features required by model M_k and f_C the features required by the configured contextual definition.

The representation $\mathcal{T}$ comprises tabular case data and the structured model and context definitions required for execution. It is worth noting, that each run selects one model collection with its metadata, one contextual definition, and one case dataset. All features required by the selected models and context must be present and mutually consistent. Section 4, Appendix C, and Supplementary Material S1 describe the corresponding files.

3.3. *Decision Integration Layer*

The Decision Integration Layer $\mathcal{OM}$ synthesizes the model outputs into the predictive score introduced in Section 3.1. For a given prioritization scenario, each model $M_k \in \mathcal{M}$ produces a prediction

$$p_{ik} = M_k(x_i), \quad p_{ik} \in [0, 1], \quad (9)$$

for case x_i . These predictions are the partial signals defined by the general architecture (Section 2.1).

To account for differences in predictive performance, each model is assigned a non-negative weight w_k , with

$$w_k \geq 0, \quad \sum_{k=1}^K w_k = 1. \quad (10)$$

The integrated decision signal for case x_i is then obtained as

$$D_i^{\text{raw}} = \Phi(p_{i1}, \dots, p_{iK}) = \sum_{k=1}^K w_k p_{ik}, \quad (11)$$

where $D_i^{\text{raw}} \in [0, 1]$ is the raw decision proposal resulting from the integration of the ADM outputs. For prioritization, it is the integrated *predictive score* passed to the Contextual Modulator.

The weights are derived from reported predictive performance. Integration uses only model outputs and does not modify the models.

This is decision-level cooperation, not a requirement for joint training. The models may differ in their training data, feature subsets, algorithms, and objectives.

3.4. Contextual Modulator

The Contextual Modulator $\mathcal{C}$ represents contextual conditions through a declarative fuzzy specification. Following the role of context in Novoa-Hernández et al. (2026) and the use of fuzzy propositions in the attrition application of Godz et al. (2026), each context definition specifies the case attributes to be considered, the corresponding fuzzy propositions, and the aggregation rules used to obtain a contextual assessment. CADEMAs-ML externalizes these definitions as JSON, so contexts and policies can change independently of the application code.

Formally, let V_r denote a case attribute and $\tilde{F}_r$ a fuzzy set representing a linguistic predicate over the universe of V_r . For case x_i , an *atomic contextual rule* evaluates the proposition “ V_r is $\tilde{F}_r$ ” as

$$\mu_r(x_i) = \mu_{\tilde{F}_r}(x_{ir}) \in [0, 1], \quad (12)$$

where x_{ir} is the value of V_r for case x_i . The supported numerical and categorical membership functions are specified in Appendix B.1.

Atomic rules may also include activation conditions. Let $m_r(x_i) \in \{0, 1\}$ indicate whether the **when** condition of rule r is satisfied for case x_i . Its effective membership is then

$$\mu_r^*(x_i) = m_r(x_i)\mu_r(x_i), \quad (13)$$

with $m_r(x_i) = 1$ for unconditional rules. This allows conditional policies to be expressed declaratively, without modifying the case data or introducing procedural logic.

Context definitions may further include *derived rules* that combine previously evaluated memberships. For a derived rule d , its membership is given by

$$\mu_d(x_i) = G_d \left(\mu_{r_1}^*(x_i), \dots, \mu_{r_q}^*(x_i) \right), \quad (14)$$

where G_d is an aggregation operator. Appendix B.3 defines the implemented operators. The contextual alignment score for case x_i is finally obtained as

$$C_i = H \left(\mu_1(x_i), \dots, \mu_L(x_i) \right), \quad (15)$$

where H is the final aggregation policy specified by the context definition.

The resulting C_i represents the degree to which case x_i satisfies the contextual conditions of the selected scenario. The tool retains every atomic and derived membership, so the assessment remains traceable to its rules.

3.4.1. Contextual Modulation

The contextual assessment C_i is subsequently used by the Contextual Modulator to transform the raw decision proposal D_i^{raw} into the final decision value used for prioritization. Within the freedom provided by the general formulation (Section 2.1), CADEMAs-ML offers weighted arithmetic, weighted geometric, and minimum operators.

Let $D_i^{\text{raw}} \in [0, 1]$ denote the integrated predictive signal and $C_i \in [0, 1]$ the contextual alignment score. The final decision value is obtained as

$$D_i^{\text{final}} = M(D_i^{\text{raw}}, C_i). \quad (16)$$

For the weighted arithmetic operator, used in previous CADEMAs applications (Godz et al., 2026; Novoa-Hernández et al., 2026), the modulation is

$$D_i^{\text{final}} = \lambda D_i^{\text{raw}} + (1 - \lambda) C_i, \quad \lambda \in [0, 1]. \quad (17)$$

CADEMAS-ML additionally implements the weighted geometric operator,

$$D_i^{\text{final}} = (D_i^{\text{raw}})^\lambda (C_i)^{1-\lambda}, \quad (18)$$

and the minimum operator,

$$D_i^{\text{final}} = \min \{ D_i^{\text{raw}}, C_i \}. \quad (19)$$

The inclusion of these alternatives follows the analysis in Novoa-Hernández et al. (2026). The geometric and minimum operators restrict compensation between a low predictive score and low contextual alignment. The parameter λ controls their relative contribution in the arithmetic and geometric cases; the minimum operator has no weight.

3.5. *Explainability Extension*

Case-level explanation exposes the intermediate artifacts produced by the integration and contextual stages. A predictive layer estimates the local influence of input features on D_i^{raw} . A contextual layer traces the fuzzy evidence underlying C_i .

For a selected case x_i , the explanation preserves the aggregation mechanism used to obtain the final decision. The predictive and contextual component values, D_i^{raw} and C_i , are reported together with the selected aggregation operator and, when applicable, its parameter λ . The explanation is additive for the linear operator and multiplicative for the geometric operator. For the minimum operator, it identifies the lower component as the limiting factor.

To explain the predictive component, CADEMAs-ML employs a model-agnostic perturbation procedure suitable for the heterogeneous and potentially ensemble-based ML models supported by the tool. For each model M_k and feature f , the local effect is estimated by comparing the original prediction with the prediction obtained after replacing the observed feature value with a cohort reference value b_f :

$$\Delta_{ikf} = M_k(x_i) - M_k \left(x_i^{(f \leftarrow b_f)} \right), \quad (20)$$

where b_f is the median of the reference cohort for numerical features and the mode for categorical features. The model-specific effects are then integrated using the same weights employed by the Decision Integration Layer:

$$\Delta_{if} = \sum_{k=1}^K w_k \Delta_{ikf}. \quad (21)$$

Positive values indicate that the observed feature value increases the integrated predictive score relative to the reference value, whereas negative values indicate a decrease. They are approximate local effects in probability points, not causal estimates. The resulting predictive explanation can therefore be represented as

$$\mathcal{E}_i^{\text{pred}} = \{(f, \Delta_{if})\}_{f \in \mathcal{F}}, \quad (22)$$

where $\mathcal{F}$ denotes the set of features considered in the perturbation analysis.

The contextual explanation traces the assessment C_i back to the fuzzy propositions and derived rules defined in the selected context. By retaining the membership values of the evaluated rules, the tool exposes the intermediate evidence underlying the contextual aggregation:

$$\mathcal{E}_i^{\text{context}} = \{(r, \mu_r^*(x_i))\}_{r \in \mathcal{R}_a} \cup \{(d, \mu_d(x_i))\}_{d \in \mathcal{R}_d}, \quad (23)$$

where $\mathcal{R}_a$ and $\mathcal{R}_d$ denote the sets of atomic and derived rules evaluated for case x_i , respectively. The use of $\mu_r^*(x_i)$ for atomic rules accounts for possible activation conditions. The retained values connect C_i to the conditions satisfied by the case.

The two explanation layers are subsequently combined into a deterministic natural-language representation. Formally, the explanation for case x_i can be represented as

$$E_i = \Psi \left(D_i^{\text{final}}, D_i^{\text{raw}}, C_i, \mathcal{E}_i^{\text{pred}}, \mathcal{E}_i^{\text{context}}, \mathcal{A}_i \right), \quad (24)$$

where $\mathcal{A}_i$ denotes the selected aggregation mechanism and its parameters when applicable. The function Ψ translates this evidence into a structured account of the predictive and contextual drivers. The quantitative artifacts remain available for inspection.

3.6. Robustness Extension

As anticipated in Section 1, robustness analysis evaluates the stability of priorities across modulation configurations. Let

$$\mathcal{G}_{\text{mod}} = \{g = (o, \theta_o)\}$$

denote the set of modulation configurations explored by the analyst, where o identifies the selected aggregation operator and θ_o denotes its associated parameterization, which may be empty for parameter-free operators. Each configuration $g \in \mathcal{G}_{\text{mod}}$ induces a prioritization score vector

$$\mathbf{D}^{(g)} = (D_1^{(g)}, \dots, D_N^{(g)})$$

and an associated ranking $\pi^{(g)}$ over the cases.

Within-operator robustness compares parameterizations of one operator. For linear and weighted geometric aggregation, the analyst varies λ and compares the resulting scores and rankings. The parameter-free minimum operator can serve as a reference.

Across-operator robustness changes the aggregation mechanism while holding the predictive and contextual signals fixed. The available operators form the set

$$\mathcal{O} = \{\text{linear, geometric, minimum}\}$$

and may induce different scores and rankings. Their comparison identifies stable and operator-sensitive cases.

More generally, the tool considers the collection

$$\mathcal{R}_{\mathcal{G}} = \left\{ \pi^{(g)} : g \in \mathcal{G}_{\text{mod}} \right\}$$

of rankings induced by the explored configurations. The analyst can inspect these rankings individually or jointly and combine within-operator and across-operator comparisons.

The analysis shows how strongly the conclusions depend on the balance between predictive and contextual evidence and on the selected operator.

4. Architecture and main functionalities

4.1. *Software architecture*

4.2. *Main functionalities*

4.2.1. *Demonstration setup: Employee-attrition case study*

4.2.2. *Integrated prioritization overview*

4.2.3. *Cooperative predictive risk audit*

4.2.4. *Contextual alignment evaluation*

4.2.5. *Case-level explanation*

4.2.6. *Robustness and sensitivity analysis*

5. Discussion

This paper presented CADEMAs-ML as an executable specialization of the CADEMAs framework for cooperative, context-aware, ML-based prioritization. Its contribution is not a new attrition predictor, but an operational separation of predictive evidence, contextual evaluation, and final prioritization. This separation builds on the general architecture of Novoa-Hernández et al. (2026) and the employee-attrition instantiation of Godz et al. (2026), while adding reusable configuration, case-level explanation, and ranking-sensitivity analysis. The employee-attrition case study shows the practical consequence of that design: rankings based on cooperative predictive risk need not coincide with rankings based on contextual alignment, and the explicit combination of both can be inspected rather than treated as an opaque post-processing step.

The tool also occupies a distinct position relative to related decision-support software. Existing platforms and libraries have made important advances in exposing intermediate fuzzy or multi-criteria calculations (Chekry et al., 2024; Więckowski & Sałabun, 2024), while hybrid approaches have combined prediction, ranking, expert knowledge, and explainability in application-specific pipelines (Özkurt et al., 2025). CADEMAs-ML complements these contributions by organizing the workflow around cooperating ADM units and by keeping their integrated output separate from the context in which a decision is made. The *Models*, *Context*, and *Overview* views preserve this distinction; the *Explain* and *Robustness* views then support case-level interpretation and policy-sensitivity analysis. These additions extend the static prioritization workflow considered in earlier CADEMAs work.

For practitioners, the main benefit of this architecture is the ability to revise decision policy without embedding that policy in the predictive models themselves. When

organizational conditions change, analysts can retain the predictive layer, provide a different context specification, and examine how the resulting priorities change. The separate reporting of R_i , C_i , and $P_i(\lambda)$ makes the source of a recommendation visible, while the λ sweep identifies cases whose position is stable and cases whose priority depends strongly on the selected balance. This does not remove the need for professional judgment; rather, it provides a structured record of the predictive and policy assumptions presented to the decision-maker. Potential applications include alert triage, review queues, retention programs, and allocation of limited support or inspection resources, provided that models, contextual rules, and decision thresholds are validated for the domain concerned.

For researchers, CADEMAs-ML provides a reproducible environment in which alternative cooperation, context-representation, explanation, and robustness mechanisms can be compared within a common architecture. The current implementation uses performance-normalized weights to combine independently trained models and a convex linear rule, $P_i(\lambda) = \lambda R_i + (1 - \lambda)C_i$, to integrate predictive risk and context. This final rule is compensatory: sufficiently strong evidence in one component can offset weaker evidence in the other. Recent work on aggregation semantics for CADEMAs (Novoa-Hernández et al., 2026) motivates extending the software with operator families that express different compensation requirements, including conjunctive or veto-like behavior. Making the aggregation semantics configurable would allow the software to distinguish settings in which compensation is acceptable from those in which a minimum level of predictive or contextual support is required.

A related direction is to move cooperation upstream, from post-hoc output aggregation to model development. At present, the predictive ADM units are trained independently and cooperation begins when their probabilities are weighted. Future versions could investigate diversity-aware model selection, theory-informed learning constraints, or multi-objective procedures that balance predictive performance, calibration, and complementarity. The aim would not be to encourage disagreement for its own sake, but to obtain models that remain individually reliable while contributing non-redundant evidence from different variables, data sources, or perspectives on the problem. Measures of conditional competence and redundancy could then inform both training and aggregation, providing a more principled trade-off between the performance of individual ADM units and the informational value of their cooperation.

Several limitations delimit these implications. The ADM units currently supported at run time are H2O MOJO classifiers; the broader CADEMAs notion of heterogeneous units, including optimization models, simulators, or rule-based systems, is not yet implemented. Local ML attributions are model-agnostic perturbations around a cohort baseline and should not be read as causal effects or as substitutes for model-specific explanation methods when model internals are available. The attrition case study uses a 20-case illustrative subset and is neither a predictive-performance study nor evidence

for general human-resources policy. Moreover, context specifications require domain-expert elicitation, and the interface does not yet provide versioned approval workflows or immutable audit trails for production governance. Future development should therefore combine broader ADM support and richer aggregation semantics with empirical validation in additional domains, stronger context governance, and evaluation with the practitioners who would interpret and act on the resulting priorities.

Funding

This research was supported by the project “Study, Analysis and Evaluation of Cooperative Automated Decision-Making Systems (CADEMAS)”, PID2023-146575NB-I00, funded by MCIU/AEI/10.13039/501100011033 and by FEDER, EU.

Disclosure statement

The authors report there are no competing interests to declare.

Declaration of generative AI use

Generative AI tools were used to assist with language editing and manuscript formatting. The authors reviewed and take full responsibility for the content of this article.

Data availability statement

The case study uses a 20-record subset of the public IBM HR Analytics Employee Attrition and Performance dataset (Subhash, 2017). Appendix A describes the subset. The data, metadata, models, and contexts are distributed with the software (Section 5).

Software availability

CADEMAS-ML is available at <https://github.com/pnovo/cademas-app> under an MIT license (Novoa-Hernández & Pelta, 2026). An archival snapshot will be deposited in Zenodo at the time of submission; the corresponding DOI will be inserted when it is available.

Supplementary material

Supplementary Material S1 accompanies this article and provides the complete specification of the four application inputs, field-level constraints, cross-file consistency requirements, and minimal configuration examples.

Author contributions

Pavel Novoa-Hernández: conceptualization, methodology, software, validation, writing – original draft, visualization. David A. Pelta: conceptualization, methodology, supervision, writing – review and editing.

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Appendix A. Attrition example dataset and model features

The 20-case attrition example distributed with CADEMAs-ML contains 36 columns: `Case_ID`, 34 predictor variables, and the `attrition` outcome label. They describe demographic, professional, compensation, performance, and organizational characteristics from the dataset used by Godz et al. (2026). Five variables (`age`, `gender`, `department`, `job_role`, and `job_level`) are shared by all four MOJO models. Two additional

columns (`employee_count` and `employee_number`) are retained in the CSV file but are not used by any model in the example bundle. Each MOJO selects its variables from the full case matrix according to the model-definition schema.

Table A1 summarizes the feature assignment. The Human Resources model reads `over18` from the CSV file; the corresponding field is named `over_18` in the model metadata.

Table A1. Predictor variables in the attrition example and the MOJO models that use each one. M_1 – M_4 denote Culture and Wellbeing, Finance, Operations and Development, and Human Resources, respectively.

Predictor	M_1 Culture & Wellbeing	M_2 Finance	M_3 Ops. & Dev.	M_4 Human Resources
age	✓	✓	✓	✓
business_travel	✓			
daily_rate		✓		
department	✓	✓	✓	✓
distance_from_home				✓
education				✓
education_field				✓
employee_count				
employee_number				
environment_satisfaction	✓			
gender	✓	✓	✓	✓
hourly_rate		✓		
job_involvement	✓			
job_level	✓	✓	✓	✓
job_role	✓	✓	✓	✓
job_satisfaction	✓			
marital_status				✓
monthly_income		✓		
monthly_rate		✓		
num_companies_worked			✓	
over18				✓
over_time	✓			
percent_salary_hike		✓		
performance_rating			✓	
relationship_satisfaction	✓			
standard_hours				✓
stock_option_level		✓		
total_working_years			✓	
training_times_last_year			✓	
work_life_balance	✓			
years_at_company			✓	
years_in_current_role			✓	
years_since_last_promotion			✓	
years_with_curr_manager			✓	
Total features	12	11	13	11

Appendix B. Context specification for the *Digital / Technological Transformation* scenario

The example repository includes two context configurations, but the walkthrough focuses on *Digital / Technological Transformation*. This appendix instantiates Equi-

tions (12)–(15) in closed form for that scenario. The equivalent JSON specification is distributed with CADEMAs-ML¹. The policy is illustrative and remains separate from the application code.

B.1. *Generic membership functions*

Each numerical atomic rule is evaluated through one of four parametric membership functions. For an input value x , the *linear increasing* function with breakpoints (a, b) , $a < b$, is

$$\mu_{a,b}^{\uparrow}(x) = \begin{cases} 0, & x \leq a, \\ \frac{x-a}{b-a}, & a < x < b, \\ 1, & x \geq b, \end{cases} \quad (\text{B1})$$

and the *linear decreasing* function with breakpoints (a, b) is its mirror image

$$\mu_{a,b}^{\downarrow}(x) = 1 - \mu_{a,b}^{\uparrow}(x) = \begin{cases} 1, & x \leq a, \\ \frac{b-x}{b-a}, & a < x < b, \\ 0, & x \geq b. \end{cases} \quad (\text{B2})$$

The *triangular* function with support (a, c) and peak b , $a < b < c$, is

$$\text{tri}_{a,b,c}(x) = \begin{cases} \frac{x-a}{b-a}, & a < x \leq b, \\ \frac{c-x}{c-b}, & b < x < c, \\ 0, & \text{otherwise,} \end{cases} \quad (\text{B3})$$

and the *trapezoidal* function with feet (a, d) and shoulders (b, c) , $a < b \leq c < d$, is

$$\Pi_{a,b,c,d}(x) = \begin{cases} \frac{x-a}{b-a}, & a < x < b, \\ 1, & b \leq x \leq c, \\ \frac{d-x}{d-c}, & c < x < d, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{B4})$$

¹https://github.com/pnovo/cademas-app/blob/main/example_attrition/context/context_digital_transformation.json.

Categorical attributes are handled by a direct map $\kappa : \mathcal{S} \rightarrow [0, 1]$ that assigns a degree to each symbolic value $s \in \mathcal{S}$, with unspecified categories defaulting to 0.

B.2. Atomic rules

Seven atomic rules evaluate recent training intensity, education level, company tenure, tenure in the current role, age in the investment window, accumulated experience, and role adaptability to technological change. For case x_i , the corresponding memberships are

$$\mu_{\text{train}}(x_i) = \mu_{1,6}^{\uparrow}(\text{training_times_last_year}_i), \quad (\text{B5})$$

$$\mu_{\text{edu}}(x_i) = \mu_{2,5}^{\uparrow}(\text{education}_i), \quad (\text{B6})$$

$$\mu_{\text{yrsCo}}(x_i) = \text{tri}_{3,10,20}(\text{years_at_company}_i), \quad (\text{B7})$$

$$\mu_{\text{yrsRole}}(x_i) = \text{tri}_{1,6,11}(\text{years_in_current_role}_i), \quad (\text{B8})$$

$$\mu_{\text{age}}(x_i) = \Pi_{22,28,50,57}(\text{age}_i), \quad (\text{B9})$$

$$\mu_{\text{exp}}(x_i) = \mu_{2,10}^{\uparrow}(\text{total_working_years}_i), \quad (\text{B10})$$

$$\mu_{\text{role}}(x_i) = \kappa_{\text{role}}(\text{job_role}_i), \quad (\text{B11})$$

where the role-adaptability map assigns expert scores to each job role:

$$\kappa_{\text{role}}(s) = \begin{cases} 0.9, & s \in \{\text{Research Scientist, Research Director}\}, \\ 0.8, & s = \text{Manager}, \\ 0.7, & s \in \{\text{Manufacturing Director, Healthcare Representative}\}, \\ 0.6, & s = \text{Sales Executive}, \\ 0.5, & s = \text{Human Resources}, \\ 0.4, & s \in \{\text{Laboratory Technician, Sales Representative}\}. \end{cases} \quad (\text{B12})$$

For example, Equation (B9) assigns full membership to ages from 28 to 50. Membership decreases linearly to zero at 22 and 57.

B.3. Derived rules and final aggregation

The aggregation operators of Equation (14), applied to a vector $\mathbf{u} = (u_1, \dots, u_q)$, are

$$\text{AND}(\mathbf{u}) = \min_j u_j, \quad \text{OR}(\mathbf{u}) = \max_j u_j, \quad \text{PROD}(\mathbf{u}) = \prod_{j=1}^q u_j, \quad \text{AVG}(\mathbf{u}) = \frac{1}{q} \sum_{j=1}^q u_j. \quad (\text{B13})$$

With these operators, three derived criteria combine the atomic memberships: *upskilling capacity* (training and education), *career scalability* (company and role tenure), and *investment horizon* (age window and experience). Formally,

$$\mu_{\text{upskill}}(x_i) = \text{AVG}(\mu_{\text{train}}(x_i), \mu_{\text{edu}}(x_i)) = \frac{1}{2}(\mu_{\text{train}}(x_i) + \mu_{\text{edu}}(x_i)), \quad (\text{B14})$$

$$\mu_{\text{career}}(x_i) = \text{PROD}(\mu_{\text{yrsCo}}(x_i), \mu_{\text{yrsRole}}(x_i)) = \mu_{\text{yrsCo}}(x_i) \mu_{\text{yrsRole}}(x_i), \quad (\text{B15})$$

$$\mu_{\text{invest}}(x_i) = \text{PROD}(\mu_{\text{age}}(x_i), \mu_{\text{exp}}(x_i)) = \mu_{\text{age}}(x_i) \mu_{\text{exp}}(x_i). \quad (\text{B16})$$

The operators encode different relationships. In *upskilling capacity*, AVG allows stronger training to compensate for weaker education. The PROD operator is conjunctive, so a near-zero input reduces *career scalability* or *investment horizon* toward zero. When no aggregation override is supplied, the JSON logic tree combines four criteria: upskilling, role adaptability, career scalability, and investment horizon.

In the reported demonstration, the *average* operator is selected explicitly in the user interface. It overrides the logic tree and averages the numeric atomic and derived memberships in the audit matrix. Let

$$\mathcal{S} = \{\text{train, edu, yrsCo, yrsRole, age, exp, role, upskill, career, invest}\}.$$

The context-alignment score used in the figures and numerical results is therefore

$$C_i = \frac{1}{|\mathcal{S}|} \sum_{r \in \mathcal{S}} \mu_r(x_i) = \frac{1}{10} \sum_{r \in \mathcal{S}} \mu_r(x_i). \quad (\text{B17})$$

Equation (B17) instantiates the generic policy in Equation (15). Its evidence is available in the *Context* module and in each case-level explanation. Selecting *minimum (strict)* or *product* in the interface applies the corresponding operator to the same audit-matrix columns.

Appendix C. Input files and consistency requirements

Table C1 details the four inputs grouped by $\mathcal{IM}$ in Section 3.2. Supplementary Material S1 provides field-level constraints and examples, including the distinction between scoring fields and display or explanation metadata.

At run time, the application checks filename matching and loads the selected metric. A deployment should also validate finite metrics, membership-parameter ordering, rule references, required features, categorical values, and positive-class consistency. These checks apply to the input bundle as a whole because its four files jointly define the executable decision model.

Table C1. Summary of the CADEMAs-ML input contract.

Input	Required content	Cross-file constraints
Case dataset (CSV)	One row per case and columns required by the uploaded models and context rules. A Case_ID or ID column is optional; otherwise, identifiers are generated.	Column names and data types must be compatible with the MOJO schemas. Every feature referenced by a context rule must be present. Identifier columns are excluded from inference.
Model metadata (JSON)	One top-level object per model, containing a performance object. The bundled schema also records department and features ; the latter is needed for the perturbation-based explanation workflow.	Each top-level key must exactly match an uploaded MOJO filename. The metric selected for weighting must be available for every participating model and should be a finite non-negative value.
Predictive models	H2O MOJO archives in .zip format that return compatible positive-class probabilities.	Each archive must have a matching metadata entry. All models must use the same positive-class interpretation, although their predictor subsets may differ.
Context specification (JSON)	A context name, atomic rules , optional derived_rules , and an optional final logic tree. Rules identify a feature, membership type, parameters, and optionally an activation condition.	Rule names must be unique, references in derived or final rules must resolve, and parameter shapes must match the membership type. The aggregation operator selected in the interface overrides the final JSON logic for that run.